

Appendix A: exact constants and certificate census

Repository source: `proof/APPENDIX_A_EXACT_CONSTANTS.md`

Contents

Main coefficient	2
Circumradius split	2
Lower-circumradius partition	2
Row 0	2
Row 01A	2
Row 01B	2
Row 02A	3
Row 02B	3
Row 03A1R1	3
Row 03A1R2	3
Row 03A1R3	3
Row 03A1R4	3
Row 03A2	4
Row 03B01	4
Row 03B02	4
Lower-branch summary	4
Near-Jung fan census	4
Near-Jung exact inequalities	4
Minimum fan margin	4
Total-deficiency upper bound	5
Determinant-squared lower bound	5
Reference-fan coefficient	5
Active-bound census	5
Dominant-edge coordinate-width inputs	5
Core, caps, and universal handoff	5
Forced-core lower bound	5
Three-pair cap lower bound	5
Near-Jung volume lower bound	5
Near-Jung margin over theorem upper enclosure	5
Coordinate-axis cap pairs	6
Axis 0	6
Axis 1	6
Axis 2	6
Cross-axis disjointness	6
Ellipsoid radius-squared upper bound	6
Overlap-norm-squared lower bound	6
Strict separation gap	6
Immutable object fingerprints	6
Reference handoff certificate	6
Reference near-Jung fan	6
Adjusted universal fan	7
Claim boundary copied from the certificate	7
Certified inside this package	7

Not established	7
Verification	7

Document status: generated from the authoritative machine-readable certificate.

Certificate schema: minvol.universal_contact_bound.v1

Source: certificate/certificates/universal_contact_bound_certificate.json

This file is generated by `proof/generate_exact_constants.py`. Do not edit it by hand. The exact verifier and certificate JSON remain the proof authority; this appendix is a human-readable projection of selected theorem-critical fields.

Very large exact fractions are identified by their JSON field and numerator/denominator digit counts rather than printed across the page. The full integers remain in the checksummed certificate. Decimal values marked “lower” are rounded downward from the corresponding exact rational; lower-row volume decimals also use the certificate’s exact rational lower bound for π .

Main coefficient

For every convex body $K \subset \mathbb{R}^3$ of constant width $d > 0$, the package represents the proposed inequality

$$\text{Vol}(K) \geq \frac{26220940089713\pi}{20000000000000} d^3 = 0.4118775637803022742481789259828398710025748462041567278 \dots d^3.$$

- Exact coefficient of π : 26220940089713/20000000000000
- Certified decimal lower enclosure: 0.4118775637803022742481789259828398710025748462041567278
- Strict gain over the predecessor certificate: 0.0000202057087485067278752862995149697425463581710298836

Circumradius split

Field	Exact value	Decimal/rendered value
Split R_*	305695601/500000000	0.61139120200000000000
Lower branch	[1/2, 305695601/500000000]	-
Near-Jung branch	[305695601/500000000, sqrt(6)/4]	-
Gap-free flag	True	-

Lower-circumradius partition

Each row below records the exact interval, tangency radius, grid-step counts, and coefficient of π . The displayed volume lower is a uniform 18-place certified decimal lower bound computed from the exact coefficient and the certificate’s rational lower bound for π .

Row 0

- Circumradius interval: [1/2, 3/5]
- Tangency radius s_0 : 37/80
- Propagation steps: 0 positive; 0 negative
- Exact coefficient of π : 76439431/575385750
- Certified volume lower: 0.417357494296971905

Row 01A

- Circumradius interval: [3/5, 6047/10000]
- Tangency radius s_0 : 959367/2000000
- Propagation steps: 7449 positive; 15746 negative
- Exact coefficient of π : 8291913333733/6250000000000
- Certified volume lower: 0.416797024215341508

Row 01B

- Circumradius interval: [6047/10000, 3047/5000]

- Tangency radius s_0 : 4819069/10000000
- Propagation steps: 7304 positive; 16448 negative
- Exact coefficient of π : 65638902944279/500000000000000
- Certified volume lower: 0.412421390558880711

Row 02A

- Circumradius interval: [3047/5000, 61019/100000]
- Tangency radius s_0 : 3943/8000
- Propagation steps: 6634 positive; 17927 negative
- Exact coefficient of π : 131363601757541/1000000000000000
- Certified volume lower: 0.412690926230586057

Row 02B

- Circumradius interval: [61019/100000, 30549/50000]
- Tangency radius s_0 : 123513/250000
- Propagation steps: 6557 positive; 18107 negative
- Exact coefficient of π : 8196358576301/625000000000000
- Certified volume lower: 0.411993918231918694

Row 03A1R1

- Circumradius interval: [30549/50000, 30553/50000]
- Tangency radius s_0 : 1242629/2500000
- Propagation steps: 12739 positive; 36890 negative
- Exact coefficient of π : 32797449039141/250000000000000
- Certified volume lower: 0.412144899831403954
- Margin over 03B02 as a coefficient of π : 85095707999/1000000000000000
- Angular denominator: 76800
- Radial denominator: 40000000000

Row 03A1R2

- Circumradius interval: [30553/50000, 30557/50000]
- Tangency radius s_0 : 310753/625000
- Propagation steps: 12719 positive; 36931 negative
- Exact coefficient of π : 4098989998007/312500000000000
- Certified volume lower: 0.412075419676058637
- Margin over 03B02 as a coefficient of π : 62979487659/1000000000000000
- Angular denominator: 76800
- Radial denominator: 40000000000

Row 03A1R3

- Circumradius interval: [30557/50000, 30561/50000]
- Tangency radius s_0 : 2486799/5000000
- Propagation steps: 12699 positive; 36973 negative
- Exact coefficient of π : 26229116212741/200000000000000
- Certified volume lower: 0.412005994020500329
- Margin over 03B02 as a coefficient of π : 2044030757/500000000000000
- Angular denominator: 76800
- Radial denominator: 40000000000

Row 03A1R4

- Circumradius interval: [30561/50000, 6113/10000]
- Tangency radius s_0 : 4975169/10000000
- Propagation steps: 12679 positive; 37015 negative

- Exact coefficient of π : 131123499722913/1000000000000000
- Certified volume lower: 0.411936623442486770
- Margin over 03B02 as a coefficient of π : 4699818587/2500000000000000
- Angular denominator: 76800
- Radial denominator: 40000000000

Row 03A2

- Circumradius interval: [6113/10000, 244541/400000]
- Tangency radius s_0 : 1987/4000
- Propagation steps: 6397 positive; 18448 negative
- Exact coefficient of π : 131111451030361/1000000000000000
- Certified volume lower: 0.411898771358480044

Row 03B01

- Circumradius interval: [244541/400000, 76421809/125000000]
- Tangency radius s_0 : 4979241/10000000
- Propagation steps: 6313 positive; 18555 negative
- Exact coefficient of π : 32777190301709/2500000000000000
- Certified volume lower: 0.411890321028654451

Row 03B02

- Circumradius interval: [76421809/125000000, 305695601/500000000]
- Tangency radius s_0 : 2489911/5000000
- Propagation steps: 6309 positive; 18562 negative
- Exact coefficient of π : 26220940089713/2000000000000000
- Certified volume lower: 0.411877563780302274

Lower-branch summary

- Unique bottleneck: 03B02
- Weakest exact coefficient of π : 26220940089713/2000000000000000
- Refined replacement cells: 4
- Refined angular denominator: 76800
- Refined radial denominator: 40000000000
- Minimum exact coefficient margin over 03B02: 4699818587/2500000000000000
- Replaced predecessor row: 03A1

Near-Jung fan census

Quantity	Integer value
Fan vertices	98306
Fan triangular faces	196608
Fan edges	294912
Stabilizer orbits	24833
Changed vertices	18229
Minimum cone determinant	512

Near-Jung exact inequalities**Minimum fan margin**

- Exact: certificate field `near_Jung_branch.minimum_margin` (42-digit numerator / 52-digit denominator)
- Certified decimal lower: 0.000000000288890420709285

Total-deficiency upper bound

- Exact: 600399058497598/30949800470751201
- Decimal rendering: 0.019399125337334536866158

Determinant-squared lower bound

- Exact: certificate field `near_Jung_branch.determinant_squared_lower` (33-digit numerator / 33-digit denominator)
- Certified decimal lower: 0.990112274299405924162539

Reference-fan coefficient

- Exact: certificate field `near_Jung_branch.reference_fan_coefficient` (51-digit numerator / 51-digit denominator)
- Certified decimal lower: 0.585182474560699336095483

Active-bound census

Bound class	Count
convex	4
dominant	81663
edge	0
tie	16639

Dominant-edge coordinate-width inputs

- Distinguished edge: `e01`
- Axis 0 diagonal bound: $H_{00} \leq 1 + S/2$
- Axis 1 diagonal bound: $H_{11} \leq 1 + S/6$
- Axis 2 diagonal bound: $H_{22} \leq 1 + S/6$
- Positive-definite inverse argument: For positive-definite H , Cauchy-Schwarz gives $1 \leq H_{jj}^*(H^{-1})_{jj}$.

Core, caps, and universal handoff**Forced-core lower bound**

- Exact: certificate field `near_Jung_branch.core_volume_lower` (201-digit numerator / 202-digit denominator)
- Certified decimal lower: 0.411735710312777173548969

Three-pair cap lower bound

- Exact: certificate field `near_Jung_branch.total_cap_lower` (4111-digit numerator / 4114-digit denominator)
- Certified decimal lower: 0.000143960902055258183828

Near-Jung volume lower bound

- Exact: certificate field `near_Jung_branch.volume_lower` (4114-digit numerator / 4114-digit denominator)
- Certified decimal lower: 0.411879671214832431732797

Near-Jung margin over theorem upper enclosure

- Exact: certificate field `near_Jung_branch.margin_over_theorem_upper` (4152-digit numerator / 4157-digit denominator)
- Certified decimal lower: 0.000002107434530157484619

Coordinate-axis cap pairs

Axis 0

- Coordinate-width-squared lower: 30949800470751201/3125000000000000
- Coordinate-width lower (certified decimal): 0.995185216461759158
- Pair-cap lower: certificate field `near_Jung_branch.cap_pairs[0].pair_cap_lower` (2060-digit numerator / 2064-digit denominator)
- Pair-cap lower (certified decimal): 0.000050275601362298174224
- Section hull sizes, positive/negative: 110/84

Axis 1

- Coordinate-width-squared lower: 92849401412253603/93149600941502402
- Coordinate-width lower (certified decimal): 0.998387315752931109
- Pair-cap lower: certificate field `near_Jung_branch.cap_pairs[1].pair_cap_lower` (2617-digit numerator / 2621-digit denominator)
- Pair-cap lower (certified decimal): 0.000046842650346480004801
- Section hull sizes, positive/negative: 102/76

Axis 2

- Coordinate-width-squared lower: 92849401412253603/93149600941502402
- Coordinate-width lower (certified decimal): 0.998387315752931109
- Pair-cap lower: certificate field `near_Jung_branch.cap_pairs[2].pair_cap_lower` (2617-digit numerator / 2621-digit denominator)
- Pair-cap lower (certified decimal): 0.000046842650346480004801
- Section hull sizes, positive/negative: 102/76

Cross-axis disjointness

Ellipsoid radius-squared upper bound

- Exact: 92849401412253603/30349401412253603
- Decimal upper rendering: 3.059348688661963485

Overlap-norm-squared lower bound

- Exact: certificate field `near_Jung_branch.cross_axis_disjointness.overlap_norm_squared_lower` (57-digit numerator / 56-digit denominator)
- Certified decimal lower: 3.697226338412901248

Strict separation gap

- Exact: certificate field `near_Jung_branch.cross_axis_disjointness.strict_gap` (72-digit numerator / 73-digit denominator)
- Certified decimal lower: 0.637877649750937762840064

Immutable object fingerprints

Reference handoff certificate

```
36a6d5cc145fdda500b3cfa8f77a784e6e29860efc44196b05684ee4c348b7af
```

Reference near-Jung fan

```
dcd9c4945cad4960f0a1a933d34b272190bb21d351e03eec3c2d4a708ed26200
```

Adjusted universal fan

```
7ad091a639b46f55e87672d38687840d8e1196df9700224816743974334a1ed7
```

Claim boundary copied from the certificate

Certified inside this package

- Gap-free lower circumradius partition through O3B02.
- Four exact doubled-angular-grid cells replacing O3A1.
- Exact dominant-edge coordinate-width sharpening for all three coordinate axes.
- Exact adjusted 98,306-ray near-Jung fan at the split.
- Three separately optimized antipodal cap pairs and exact six-cap disjointness.
- Strict exact near-Jung handoff margin and gain over the predecessor certificate.

Not established

- External independent reproduction or peer review.
- Equality, rigidity, or sharpness.
- Identification of a minimizing body or Meissner extremality.
- The universal support-box inequality.

Verification

Regenerate this appendix deliberately with:

```
python3 proof/generate_exact_constants.py
```

Verify that it is current without modifying it with:

```
python3 proof/generate_exact_constants.py --check
```